Candidate Discovery Engine (CDE) - Multi-Factor Scoring

Asset Universe (106 Active Assets)

Hard eligibility filter: is_active, >=10 days price history, not already in portfolio

Expected Return

ML Prediction (annualized)

w_ret = 0.25 +(risk-3)x0.1

Volatility (Risk)

Historical Vol (annualized)

w_vol = 0.25 -(risk-3)x0.1

Momentum (Trend)

90-day cum return

w_mom = 0.15 +(risk-3)x0.05

Diversif. Benefit

1 - corr(asset, portfolio)

w_div = 0.20 +(3-risk)x0.05

Fundam. Quality

ROE, margins, debt, FCF

w_fund=0.15 (fixed)

Signal Normalization -> [0, 1]

n_ret = clamp(E[R] / 0.5) | n_vol = clamp(1 - Vol/0.5) | n_mom = clamp((mom + 0.2) / 0.4) | n_div = clamp(div/2)

Weighted Composite Score

*Score = Sum (w_i / Sum_w) * n_i * 100*

Risk-Adaptive Weight Distribution

Risk Level	w_ret	w_vol	w_div	w_mom	w_fund	Profile
1 (Conservative)	0.05	0.45	0.30	0.05	0.15	Safety-First
3 (Moderate)	0.25	0.25	0.20	0.15	0.15	Balanced
5 (Aggressive)	0.45	0.05	0.10	0.25	0.15	Growth-Seeking

Top-N Ranked Candidates -> APDE Optimizer